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缺题，可补交

9.5+1

第四次作业

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13.

由 $\vec{B}=\frac{\mu_0 q\vec{v}\times\vec{r}}{4\pi r^3}$ 得 q_1 在 q_2 处产生 $B_{21}=\frac{\mu_0 q_1 v_1 d}{4\pi d^3}=\frac{\mu_0 q_1 v_1}{4\pi d^2}$

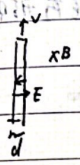
垂直纸面向

$\vec{B}_{12}=\vec{0}$

$\therefore\ F_{21}=v_2 q_2 B_{21}=\frac{\mu_0 q_1 q_2 v_1 v_2}{4\pi d^2}$，方向水平向右

$F_{12}=0$

14.



平衡时 $Eq=vBq$

$U=Ed$

$\Rightarrow\ v=\frac{E}{B}=\frac{U}{Bd}=37\ m/s$

系列化习题

10.

(1) M处：圆柱面上电流不贡献磁场

$\vec{B}_M=-\frac{\mu_0 \alpha}{2}\hat{y}$

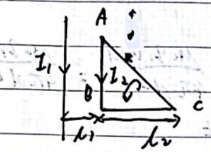
而 $\vec{B}_N=-\frac{\mu_0 \alpha}{2}\hat{y}+\frac{2}{3}\mu_0\alpha\hat{x}$

(2) 圆柱面处 磁感应强度 $\vec{B}=-\frac{\mu_0\alpha}{2}\hat{y}$

单位长度受力 $\vec{f}=-\alpha B\alpha R\hat{y}$

$=-\alpha^2\pi R\hat{x}$

11.



由于 I_2 不变

$$dA = -I_2 \xi dt = +I_2 d\phi$$

$$A = +I_2 \int d\phi = -I_2(\phi - \phi')$$

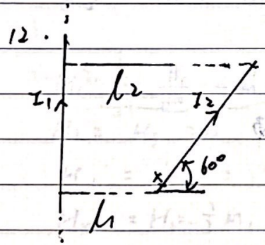
由规定的方向正负，知 ϕ、ϕ' 等价于上面正方形中的磁通

$$\phi - \phi' = \int_{l_1}^{l_1+l_2} l_2 \cdot \frac{\mu_0 I_1}{2\pi r} dr$$

$$= l_2 \frac{\mu_0 I_1}{2\pi} \ln \frac{l_1+l_2}{l_1}.$$

$$A = -\frac{\mu_0 I_1 I_2}{2\pi} l_2 \ln \frac{l_1+l_2}{l_1}.$$

12.



$$B_r = \frac{\mu_0 I_1}{2\pi r}$$

$$dF = I_2 B_r dl$$

$$= I_2 B_r \frac{dr}{\cos 60°}$$

$$= \frac{\mu_0 I_1}{\pi} \frac{dr}{r}$$

$$F = \int_{l_1}^{l_2} \frac{\mu_0 I_1 I_2}{\pi r} dr = \frac{\mu_0 I_1 I_2}{\pi} \ln \frac{l_2}{l_1}.$$

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14. $\circlearrowright \omega$

磁矩：$\mu = \int d\mu$

在 $r \to r+dr$ 内的磁矩：

$d\mu = \pi r^2 \cdot dI = \pi r^2 \frac{dq}{\frac{2\pi}{\omega}} = \omega \cdot \sigma_e \pi r^3 dr$.

σ_e 为电荷面密度 $\frac{q}{\pi(R_2^2 - R_1^2)}$

$\mu = \int d\mu = \int_{R_1}^{R_2} \omega \sigma_e \pi r^3 dr$

$= \frac{1}{4}\omega q (R_1^2 + R_2^2)$

$\vec{M} = \vec{\mu} \times \vec{B} = \frac{1}{4}\omega q B \sin\alpha (R_1^2 + R_2^2) \vec{e_n}$

16.

(1) 设 $n = \frac{N}{R_2 - R_1}$

$m = \int_{R_1}^{R_2} I \cdot \pi r^2 n dr = \frac{1}{3}\pi (R_2^2 + R_2 R_1 + R_1^2) NI$

(2) $M_m = |\vec{\mu} \times \vec{B}| = \frac{1}{3}\pi BNI (R_2^2 + R_2 R_1 + R_1^2)$

方向向上

(3) $dA = M_m d\theta$

$M_m(\theta) = \frac{1}{3}\pi BNI (R_2^2 + R_2 R_1 + R_1^2) \cos\theta$

$A = \int dA = \int_0^{\frac{\pi}{2}} \frac{1}{3}\pi BNI (R_2^2 + R_2 R_1 + R_1^2) \cos\theta d\theta$

$= \frac{1}{3}\pi BNI (R_2^2 + R_2 R_1 + R_1^2)$

$\rho_s = \frac{2v^2 t(-a^2-2ad)}{(M^2)}$

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系列6

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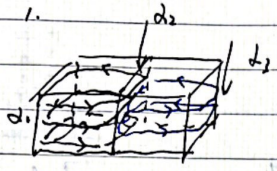
1.

(1)



(2) $\alpha_1 = M_1$, $\alpha_2 = M_1 - M_2$, $\alpha_3 = M_2$

(3)

$m = m_1 + m_2 = M_1 a \cdot a^2 + M_2 a \cdot a^2$

$(M_1 + M_2) a^3$

2.

(1) 左铜 $M_1 = \frac{B}{\mu_0 \mu_{r1}}$，右铜 $M_2 = \frac{B}{\mu_0 \mu_{r2}}$

各自方向与其对应磁场方向一致

正

左介质：$\alpha_1' = \frac{B}{\mu_0}\left(1 - \frac{1}{\mu_{r1}}\right)$，右介质：$\alpha_2' = \frac{B}{\mu_0}\left(1 - \frac{1}{\mu_{r2}}\right)$

方向均垂直向上

(2)

$I = \oint_l \left(\frac{\vec{B}}{\mu_0} - \vec{M}\right) d\vec{l} = \frac{Bl}{\mu_0 \mu_{r1}} + \frac{Bl}{\mu_0 \mu_{r2}}$

$\alpha_0 = \frac{I}{l} = \frac{B}{\mu_0}\left(\frac{1}{\mu_{r1}} + \frac{1}{\mu_{r2}}\right)$

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(续)

3.

(1) $B = 1.0\,T$

有 $H \cdot C_{环} = NI$, $H = \dfrac{B}{\mu_0 \mu_r}$

可 $H = 2\times10^4\ A/m$, $\mu_r = 39.8$

(2) $M = \dfrac{B}{\mu_0} - H$

$= 7.757\times10^{4}\ A/m$

(3) $\dfrac{M}{H} = 38.9$

(4) $\mu_r = 39.8$, $T' = \alpha \cdot C$ $MC = 3.1\times10^{5}\ A$

4.

$B = \mu_0 n I = \mu_0 M$ (磁化强度 [illegible])

$B_1 = B_2 = \mu_0 M$.

$H_1 = \dfrac{B_1}{\mu_0} - M = M$

$H_2 = 0$

5. (1)

(2) $B_1 = B_2 = B_3 = \dfrac{\mu_0 M h}{2R}$

$H_2 = \dfrac{B}{\mu_0} - M = \dfrac{Mh}{2R} - M$, $H_1 = H_3 = \dfrac{Mh}{2R}$.

(3) M 沿 y 方向 ⇔ 四大平板的磁场

$B_1 = \mu_0 M$, $B_2 = B_3 = 0$

$H_1 = \dfrac{B_1}{\mu_0} - M = 0$, $H_2 = H_3 = 0$

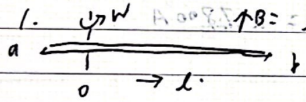
6. (1) 当环中 H 开始达到 H_c 时，翻转开始

$I=-0.5A$ 时，开始翻转，I与图中取向一致为正方向.

(2) $I_p = 0.8A$

教材 P373

1. ω $B=5\times10^{-5}T$



$$\varepsilon_{ab}=\int_a^b(\vec{v}\times\vec{B})\cdot d\vec{l}=\int_{0.1}^{0.4}\omega r B\,dr=4\pi B\cdot\frac{1}{2}(0.4^2-0.1^2)$$

$$=9.71\times10^{-5}V$$

2.

$$\varepsilon_{ab}=\int_0^R\omega r B\,dr=\frac{1}{2}(2N\pi)BR^2$$

$$=N\pi BR^2$$

(2) $M=\int_0^R IrB\,dr=\frac{1}{2}R^2IB$，方向与角速度相反

(3) 会

(4) 若认为原理性质理想，那么与上述情况有无二致

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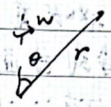
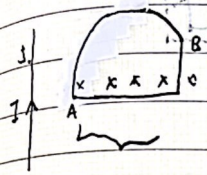
4.



$$\varepsilon = \int_0^A (\vec{v}\times\vec{B})\cdot d\vec{l} = \int_0^l \omega r\sin\theta B\sin\theta\, dr$$

$$= \frac{1}{2}\omega\sin^2\theta B l^2.$$



$$\varepsilon_{AB} = \int_R^{\frac{5}{2}R} -\frac{\mu_0 I}{2\pi r}\, dr$$

$$= -\frac{\mu_0 I}{2\pi}\ln\frac{5}{2}$$

6. 环中磁通 $\Phi_m(t) = 3\pi r^2 B(t)$

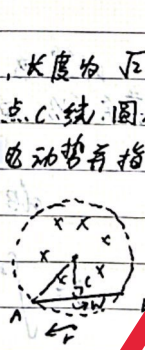
$$I = -\frac{1}{R}\frac{d\Phi}{dt} = -\frac{1}{R}3\pi r^2 \omega B_0 \cos\omega t$$

$$\therefore I_{max} = \frac{3\pi r^2}{R}\omega B_0 = 0.4\ mA.$$

补充题

1. 磁感应强度为B的匀强磁场垂直纸面向里，长度为 $\sqrt{2}R$ 的杆两端点A、B始终维持在虚线圆周上，杆中点C绕圆心O以角速度ω在纸面内逆时针旋转，求AC上的动生电动势并指明方向。

建立自然坐标系 r.



$$\vec{v}_r = \vec{\omega}\times(\vec{r_c}+\vec{r})$$

$$\varepsilon_{AC} = \int_A^C [\vec{\omega}\times(\vec{r_c}+\vec{r})]\times\vec{B}\cdot d\vec{r}$$

$$= \int_A^C (\vec{\omega}\times\vec{r_c})\times\vec{B}\cdot d\vec{r} + \int_A^C (\vec{\omega}\times\vec{r})\times\vec{B}\cdot d\vec{r}$$

$$= 0 + \int_A^C -\omega B r\, dr = \int_{\frac{1}{2}R}^{R} -\omega B r\, dr = \frac{1}{4}\omega B R^2$$

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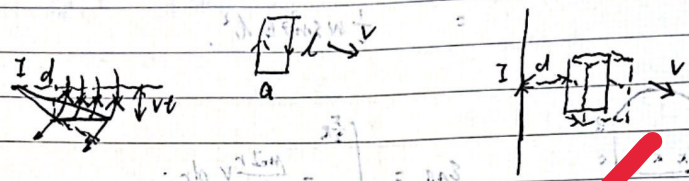
2. 一根长直导线通有恒定的电流 I，长 l 宽 a 的矩形线圈初始时与导线共面，其两对边与导线平行，然后以垂直于线圈平面的速度 v 做匀速运动，求：t 时刻线圈中的感应电动势（忽略线圈自感）



移动 t 时长后，线圈磁通 $\Phi_m(t)$

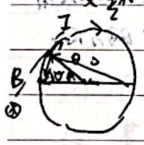
$$\Phi_m(t) = \int_{r_1}^{r_2} \frac{\mu_0 I}{2\pi r} \cdot l \cdot dr$$

而 $r_1 = \sqrt{d^2+(vt)^2}$，$r_2 = \sqrt{(d+a)^2+(vt)^2}$

$$\Rightarrow \Phi_m(t) = \frac{\mu_0 I}{2\pi} l \ln \frac{\sqrt{(d+a)^2+(vt)^2}}{\sqrt{d^2+(vt)^2}} = \frac{\mu_0 I}{4\pi} l \ln \frac{(d+a)^2+v^2t^2}{d^2+v^2t^2}$$

$$\varepsilon = -\frac{d\Phi}{dt} = \frac{\mu_0 I l}{4\pi} \frac{2v^2t(a^2+2ad)}{(d+a)^2+v^2t^2} \cdot \frac{1}{d^2+v^2t^2}$$

附加：计算电流为 I 半径为 R 的载流圆环中的一段圆弧受环其余部分电流的作用力



$$dB = \frac{\mu_0 I \cdot 2R\,d\theta}{4\pi \rho^2} \left(\sin\left(\frac{\pi}{2}-\theta\right)\right)$$

$$B = \int_0^{2\pi} \frac{\mu_0 I R \cos\theta}{4\pi \rho^2} d\theta = \int_0^{2\pi} \frac{\mu_0 I R \cos\theta\, d\theta}{4\pi (2R\cos\theta)^2}$$

$$= \frac{\mu_0 I}{16\pi R} \int_0^{2\pi} \frac{1}{\cos\theta} d\theta$$

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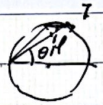

$$dF = I\, d\vec{l} \times \vec{B}$$

$$\Rightarrow \frac{dF}{dl} = I \cdot \hat{l} \times \vec{B} = I \cdot \int \hat{l} \times d\vec{B}$$

$$= I \cdot \int \hat{l} \times \frac{\mu_0 I (2R\, d\theta) \cdot l \cos\theta}{4\pi l^3}$$

$$l = 2R\cos\theta$$

$$= \frac{\mu_0 I^2}{16\pi R} \int_{-\frac{\pi}{2}+\frac{\Delta\theta}{2}}^{\frac{\pi}{2}-\frac{\Delta\theta}{2}} \frac{d\theta}{\cos\theta}$$

$$= \frac{\mu_0 I^2}{16\pi R} \left(2\,\mathrm{arctanh}\left(\cos\frac{\Delta\theta}{2}\right) \right)$$

$$= \frac{\mu_0 I^2}{16\pi R} \ln \frac{1+\cos\frac{\Delta\theta}{2}}{1-\cos\frac{\Delta\theta}{2}}$$

$$F = \frac{\mu_0 I^2 R\theta}{16\pi} \ln \frac{1+\cos\frac{\Delta\theta}{2}}{1-\cos\frac{\Delta\theta}{2}}$$

李

2023.11.18